Software Testing and Quality Assurance (STQA)

SOFTWARE REQUIREMENT SPECIFICATION

for

Online Library Management System

Name	Mayur Zope
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1. Introduction

1.1 Purpose of this Document

This document describes the complete requirements of the Online Library Management System (OLMS). It explains what the system must do, who will use it, and the conditions under which it must work. It is intended for the development team, testers, the library staff (customer), and the project guide, and serves as the base for design, coding, and test-case preparation.

1.2 Scope of this Document

OLMS is a web-based application that automates the daily work of a college library: adding and searching books, registering members, issuing and returning books, calculating fines, and generating reports. It replaces manual registers, saves time, reduces errors, and lets students search the catalogue and check their issued books online. The estimated development time is 10 weeks with a team of three members, using free and open-source tools to keep the cost low.

1.3 Overview

OLMS is a three-user system (Admin, Librarian, Student) with a central database. The Librarian manages books and transactions, the Student searches, reserves, and views records, and the Admin manages users and system settings. The remaining sections of this document cover the general description, functional and interface requirements, performance, constraints, non-functional attributes, schedule and budget, and appendices.

2. General Description

2.1 Product Perspective and Objective

OLMS is a standalone web application. The main objective is to make library operations fast, accurate, and paperless, and to give students easy access to library information at any time.

2.2 User Characteristics

User	Description	Technical Skill
Admin	Manages accounts, roles, backups, and settings.	Moderate
Librarian	Manages books, issues and returns, fines, reports.	Basic computer knowledge
Student / Member	Searches books, reserves, views history and fines.	Basic internet knowledge

2.3 Features and Benefits

- Fast book search by title, author, subject, or ISBN.
- Automatic due-date and fine calculation, avoiding manual errors.
- Online reservation and renewal of books.
- Real-time stock status and detailed reports for management.
- Reduced paperwork and quicker service for students.

2.4 General Assumptions

- All users have access to a computer or smartphone with an internet browser.
- Library books are labelled with unique ISBN or accession numbers.
- Fine rule: Rs. 2 per day after the due date of 14 days.

3. Functional Requirements

Functional requirements are listed in ranked order of priority (High, Medium, Low).

ID	Requirement	Priority
FR1	Login: The system shall authenticate Admin, Librarian, and Student using username and password.	High
FR2	Member registration: The Librarian shall add, edit, and deactivate member records.	High
FR3	Book management: The Librarian shall add, update, delete, and categorise books with ISBN, title, author, publisher, and copies.	High
FR4	Search: Users shall search books by title, author, subject, or ISBN and see availability.	High
FR5	Issue book: The system shall issue a book to a member, limit of 3 books per member, and set a due date of 14 days.	High
FR6	Return book: The system shall record returns, update stock, and calculate fine if overdue.	High
FR7	Fine calculation: Fine = overdue days x Rs. 2; the system shall store and display unpaid fines.	Medium
FR8	Reservation and renewal: Students shall reserve an unavailable book and renew an issued book once.	Medium
FR9	Notifications: The system shall send email reminders 2 days before the due date.	Medium
FR10	Reports: The system shall generate reports of issued books, overdue books, fines collected, and popular books.	Medium
FR11	Password reset: Users shall reset forgotten passwords through registered email.	Low

Sample Use Case: Issue Book

Item	Details
Actor	Librarian
Pre-condition	Librarian is logged in; member is active; book copy is available.
Main flow	1. Enter member ID. 2. Enter book ISBN. 3. System checks limit and availability. 4. System issues the book and shows the due date.
Post-condition	Stock reduced by one; transaction saved in the database.
Exception	If member has 3 books or unpaid fine, the system rejects the issue and displays a message.

4. Interface Requirements

- **User interface:** Web pages with login form, search box, forms for books and members, and report tables; responsive for desktop and mobile; consistent menus and clear error messages.
- **Hardware interface:** Standard PC, keyboard, mouse, and optional barcode scanner for reading ISBN or member card numbers.
- **Software interface:** Web server (Apache), PHP, MySQL database, and SMTP mail service for notifications.
- **Communication interface:** HTTP/HTTPS over TCP/IP; data exchanged between browser, server, and database using HTML forms and SQL queries.

5. Performance Requirements

Parameter	Requirement
Response time	Search results within 2 seconds; other pages within 3 seconds.
Capacity	Support 200 concurrent users and 100,000 book records.
Memory	Server uses at most 2 GB RAM under normal load.
Throughput	Process at least 50 issue/return transactions per minute.
Error rate	Transaction error rate below 1%.
Availability	At least 99% uptime during library working hours.

6. Design Constraints

- **Hardware limitation:** The system must run on a server with a minimum of Pentium IV class processor, 2 GB RAM, and 40 GB storage.
- **Software limitation:** Developed using PHP and MySQL; must run on Windows 7 or higher and Linux, with Chrome, Firefox, or Edge.
- **Standards:** Follow IEEE 830 for documentation and normalised database design (up to 3NF).
- **Policy constraints:** Borrowing limit, loan period, and fine rate are set by library rules and must be configurable by the Admin.
- **Time constraint:** The project must be completed within 10 weeks.

7. Non-Functional Attributes

Attribute	Description
Security	Password-protected login, role-based access, encrypted passwords (hashing), protection against SQL injection, and session timeout after 15 minutes.
Portability	Runs on any operating system through a web browser.
Reliability	Daily automatic database backup; system recovers within 1 hour after failure.
Reusability	Modules such as login, search, and reports can be reused in other projects.
Application compatibility	Compatible with all major browsers and supports import/export of data in CSV format.
Data integrity	Primary and foreign keys, validation checks, and transactions prevent duplicate or wrong data.
Scalability	Database and server can be upgraded to handle more members and books.
Usability	Easy to learn within 1 hour of training; clear menus and messages.
Maintainability	Modular code with comments so that future changes are easy.

8. Preliminary Schedule and Budget

8.1 Schedule

Phase	Activity	Duration
1	Requirement gathering and SRS	1 week
2	System design (DFD, ER diagram, UI design)	2 weeks
3	Coding and database implementation	4 weeks
4	Testing (unit, integration, system)	2 weeks
5	Deployment and user training	1 week
	Total	**10 weeks**

8.2 Budget

Item	Estimated Cost (Rs.)
Development effort (3 members x 10 weeks)	30,000
Server and hosting (1 year)	8,000
Software tools (PHP, MySQL, Apache - open source)	0
Testing and training	5,000
Miscellaneous	2,000
Total	**45,000**

9. Appendices

9.1 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
OLMS	Online Library Management System
ISBN	International Standard Book Number
FR	Functional Requirement
DFD / ER	Data Flow Diagram / Entity Relationship Diagram
SMTP	Simple Mail Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
CSV	Comma Separated Values

9.2 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Roger S. Pressman, Software Engineering: A Practitioner's Approach.
- Ian Sommerville, Software Engineering.
- Interviews with library staff and student feedback (requirement gathering).

Conclusion

Thus, the Software Requirement Specification for the Online Library Management System was prepared following the standard SRS structure, covering the introduction, general description, functional, interface, and performance requirements, design constraints, non-functional attributes, schedule and budget, and appendices.